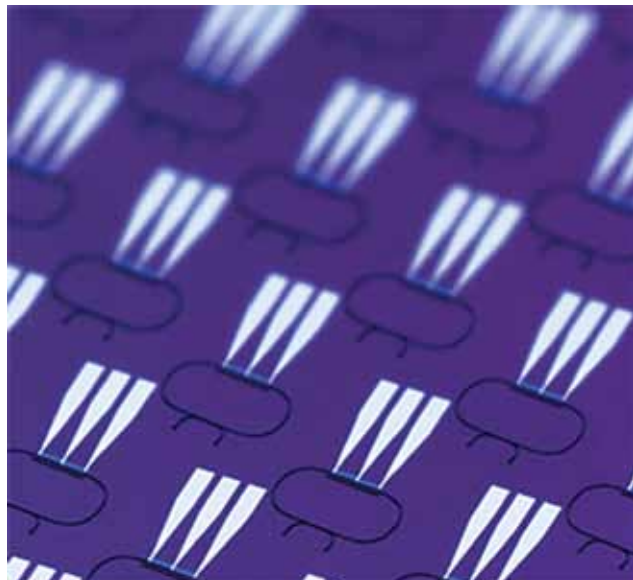


Compact IC that uses sound waves to rein in light

Besides allowing us to send data from one side of the planet to the other, laser light also helps to precisely control atoms. Currently, quantum technology is limited by size, so state-of-the-art systems would not fit on a room table, let alone a chip. But for practical use, quantum devices need to be miniaturised. Now a research team at the Illinois Quantum Information Science and Technology Centre has designed a simple, compact photonic circuit that uses sound waves to rein in light. The team's measurements show that their approach to isolation currently outperforms all previous on-chip alternatives and is optimised for compatibility with atom based sensors.

On-chip optical isolators for 780nm and 1550nm wavelengths fabricated in lithium niobate at the University of Illinois Urbana-Champaign (Credit: <https://iquist.illinois.edu>)



World's smallest brain-inspired computer being developed



Artificial intelligence (AI) has seen explosive growth in recent years, but despite major progress, the power required to run AI algorithms continues to increase. In stark contrast to this, the human brain only requires around 20W to perform more than 10 quadrillion (10,000,000,000,000,000) operations. This is 12 orders of magnitude better than modern supercomputer technologies. To bridge the gap, researchers at the Aarhus University are developing new hardware that mimics the structure of the human brain, with neurons, synapses, and neural networks, known as brain-inspired computing (BIC). On its success, the device will not only be small in size but will also be able to harvest energy directly from its surrounding environment.

Brain-inspired computing work in progress (Credit: <https://ingenioer.au.dk/en>)

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